

# Thread - 2025-03-23

<https://x.com/RiikonenMarko/status/1903696098587292129>

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Grok needs to work on its halos. Only one of the halo forms it lists is visible in the photo. Any halo-green person would have done better after just a little browsing of the internet. Pressing the "Think harder" button gave the same answer.



Identify the various ice crystal halo forms in this photo.

This photo captures a stunning display of ice crystal halos in a snowy, nighttime environment, likely taken with a fisheye lens to show a wide field of view. The bright light source at the bottom center is illuminating the scene, and the ice crystals in the atmosphere are creating various optical phenomena. Let's break down the halo forms visible in the image:

- 22° Halo:** This is the most common ice crystal halo, and while it's not prominently visible in this image due to the bright light and the fisheye distortion, there might be a faint circular ring around the light source at a radius of about 22 degrees. It's often caused by randomly oriented hexagonal ice crystals refracting light.
- Parhelic Circle:** The horizontal arc stretching across the sky, parallel to the horizon, is a parhelic circle. It's visible as a faint white band extending from the left to the right side of the image, passing through the bright light. This phenomenon occurs when light reflects off the vertical faces of ice crystals, often plate-like or columnar crystals with specific orientations.
- Sun Pillars (or Light Pillar):** The bright vertical column of light extending upward from the light source is a light pillar. This occurs when light reflects off the flat surfaces of horizontally oriented ice crystals, such as plates or dendrites, creating a vertical beam. In this case, since the light source is artificial (likely a streetlamp or similar), it's often called a light pillar rather than a sun pillar.
- Upper Tangent Arc (or possibly a Circumzenithal Arc):** The curved, rainbow-like arc near the top right of the image could be an upper tangent arc or a circumzenithal arc. An upper tangent arc forms above the 22° halo and is caused by horizontally aligned columnar ice crystals, creating a V-shaped arc. A circumzenithal arc, on the other hand, is more colorful and forms when light refracts through plate-like crystals, appearing as a bright, upside-down rainbow. Given the faint color in this arc, it's more likely a circumzenithal arc, but the low light makes it hard to confirm definitively.

5. **Parhelia (Sun Dogs)** [Think harder](#) [or I](#) [Explain circumzenithal arc](#) [or](#) [Other atmospheric optics](#) [er side of the light](#)